Person 2 Presentation Notes Models Results and Routing

Purpose: study these notes before presenting the assigned slides. Explain the ideas in your own words and use the examples when the audience is unfamiliar with AI or power grids.

What this person must understand

This section explains how the prepared data becomes a prediction, how the models are compared, what the evaluation numbers mean, and how the route planner turns a stressed link into a what-if recommendation.

Important distinction

The current website forecast endpoint uses the selected operational baseline. The saved processed artifact selects Persistence because it produced the lowest pooled chronological RMSE. Fusion GRU and DCRNN are included in the research benchmark and results pack. Do not say that DCRNN currently drives the website unless the code is changed to load the checkpoint.

Model comparison from the saved run

Model	RMSE	MAE	Accuracy	F1
Persistence	580.3 MW	375.8 MW	91.4%	81.2%
Fusion GRU	570.5 MW	400.3 MW	92.7%	85.6%
DCRNN	634.0 MW	436.2 MW	92.5%	83.8%

Slide-by-slide explanation

Slide 5 Forecasting models and system stack

Purpose: Introduce simple baselines, neural models, and the software integration.

- Persistence copies the most recent flow.
- Moving average uses the recent 24-hour mean.
- Seasonal naive uses the same hour from the previous day.
- Linear trend extrapolates the recent 24-hour slope.
- Fusion GRU and DCRNN are trained research models.

What to say: We test simple rules before trusting a more complex neural model. In this project, the website keeps Persistence as the operational baseline because it won the saved chronological comparison.

Slide 6 How the next-hour forecast is produced

Purpose: Give a numerical example and define the metrics.

- If the latest flow is 720 MW, Persistence predicts 720 MW for the next hour.
- For multiple future hours, the system appends each prediction and continues the forecast.

- The same chronological holdout is used to compare models fairly.

What to say: A baseline can be strong because hourly flow is often autocorrelated. A complex model has to earn its place by reducing error on unseen future-like data.

Slide 7 Performance evidence

Purpose: Explain the comparison chart and confusion matrix.

- Persistence RMSE: 580.3 MW.
- Fusion GRU RMSE: 570.5 MW.
- DCRNN RMSE: 634.0 MW.
- The confusion matrix uses Normal, Warning, and Critical rows and columns.

What to say: RMSE measures numeric flow error. The confusion matrix shows the type of stress-class mistake, which is more useful than accuracy alone.

Slide 8 Prescriptive routing

Purpose: Explain the road analogy and decision rules.

- Move a chosen percentage of the selected line in a simulation.
- Search for connected paths between the same endpoints.
- Reject paths without enough headroom or containing a Critical link.
- Rank survivors using distance, utilization, and estimated loss.
- Show bottlenecks and required engineering checks.

What to say: For 1,000 MW and a 25% test, the planner evaluates 250 MW on alternate paths. Purple means a candidate recommendation, not an automatic switching order.

Equations to know

$$\text{MAE} = (1/n)\ \Sigma\ |\text{prediction} - \text{actual}|$$

Average absolute flow error in MW.

$$\text{RMSE} = \sqrt{[(1/n)\ \Sigma\ (\text{prediction} - \text{actual})^2]}$$

Large errors receive more weight.

$$\text{precision} = \text{TP} / (\text{TP} + \text{FP}),\quad \text{recall} = \text{TP} / (\text{TP} + \text{FN})$$

Precision measures warning reliability; recall measures how many real warnings were caught.

$$\text{F1} = 2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$$

One score that balances precision and recall.

$$\text{moved MW} = \text{current MW} \times \text{reroute percentage} / 100$$

Example: 1,000 MW × 25% = 250 MW.

$$\text{headroom} = \text{proxy capacity} - \text{current flow}$$

Remaining screening capacity on a candidate link.

$$\text{route cost} = \text{distance} \times (1 + 2.2 \times \text{utilization} + 6 \times \text{loss percentage})$$

Lower cost is preferred after safety guardrails are applied.

Questions you should be ready for

- Why did Persistence win? Short-horizon flow can be strongly related to the latest observed value.
- Why not rely only on accuracy? Accuracy hides which class is being confused.
- Does routing change the real grid? No. It is a what-if recommendation.
- Why is MAPE not emphasized? Near-zero or sign-changing net flows make percentage error unstable.

Handoff sentence

The application can now forecast and test an action. The final section explains the advanced DCRNN evidence, the saved result figures, the novelty, and the project limits.